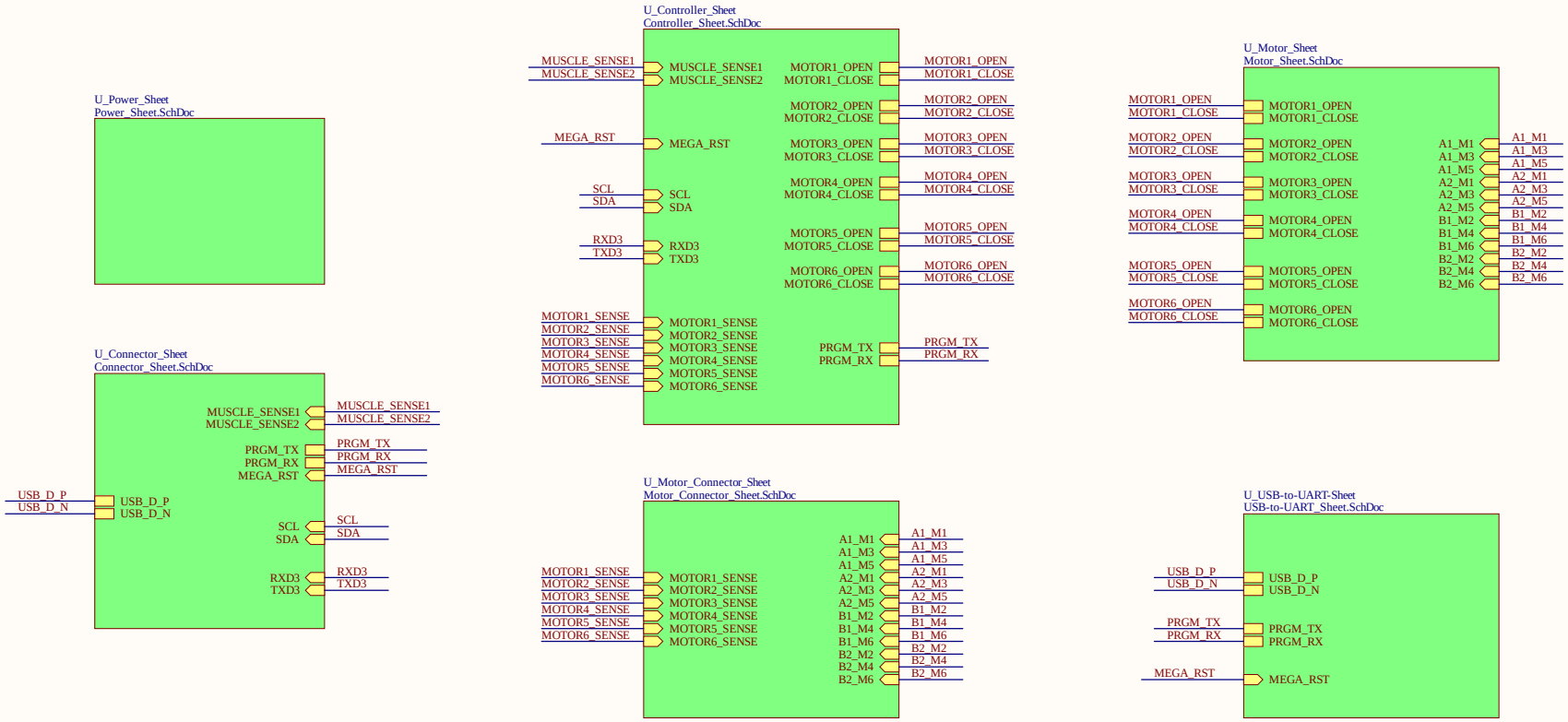
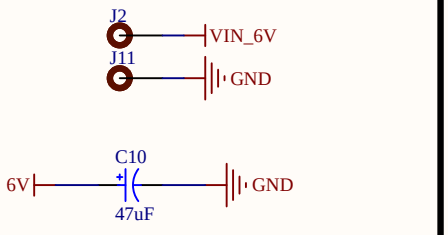
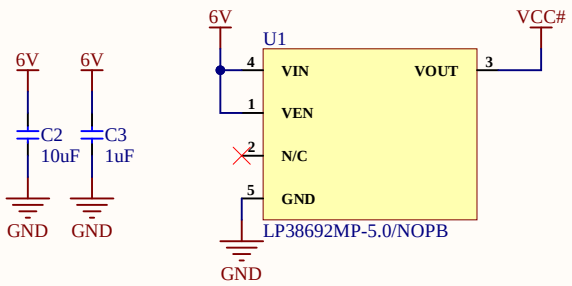

U_Power_Sheet
Power_Sheet.SchDoc
U_Connector_Sheet
Connector_Sheet.SchDoc
MUSCLE_SENSE1
MUSCLE_SENSE2
PRGM_TX
PRGM_RX
MEGA_RST
USB_D_P
USB_D_N
SCL
SDA
RXD3
TXD3
U_Controller_Sheet
Controller_Sheet.SchDoc
MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE
MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE
MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE
MOTOR1_SENSE
MOTOR2_SENSE
MOTOR3_SENSE
MOTOR4_SENSE
MOTOR5_SENSE
MOTOR6_SENSE
U_Motor_Sheet
Motor_Sheet.SchDoc
A1_M1
A1_M3
A1_M5
A2_M1
A2_M3
A2_M5
B1_M2
B1_M4
B1_M6
B2_M2
B2_M4
B2_M6
U_Motor_Connector_Sheet
Motor_Connector_Sheet.SchDoc
U_USB-to-UART-Sheet
USB-to-UART_Sheet.SchDoc

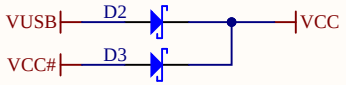
POWER INPUTS



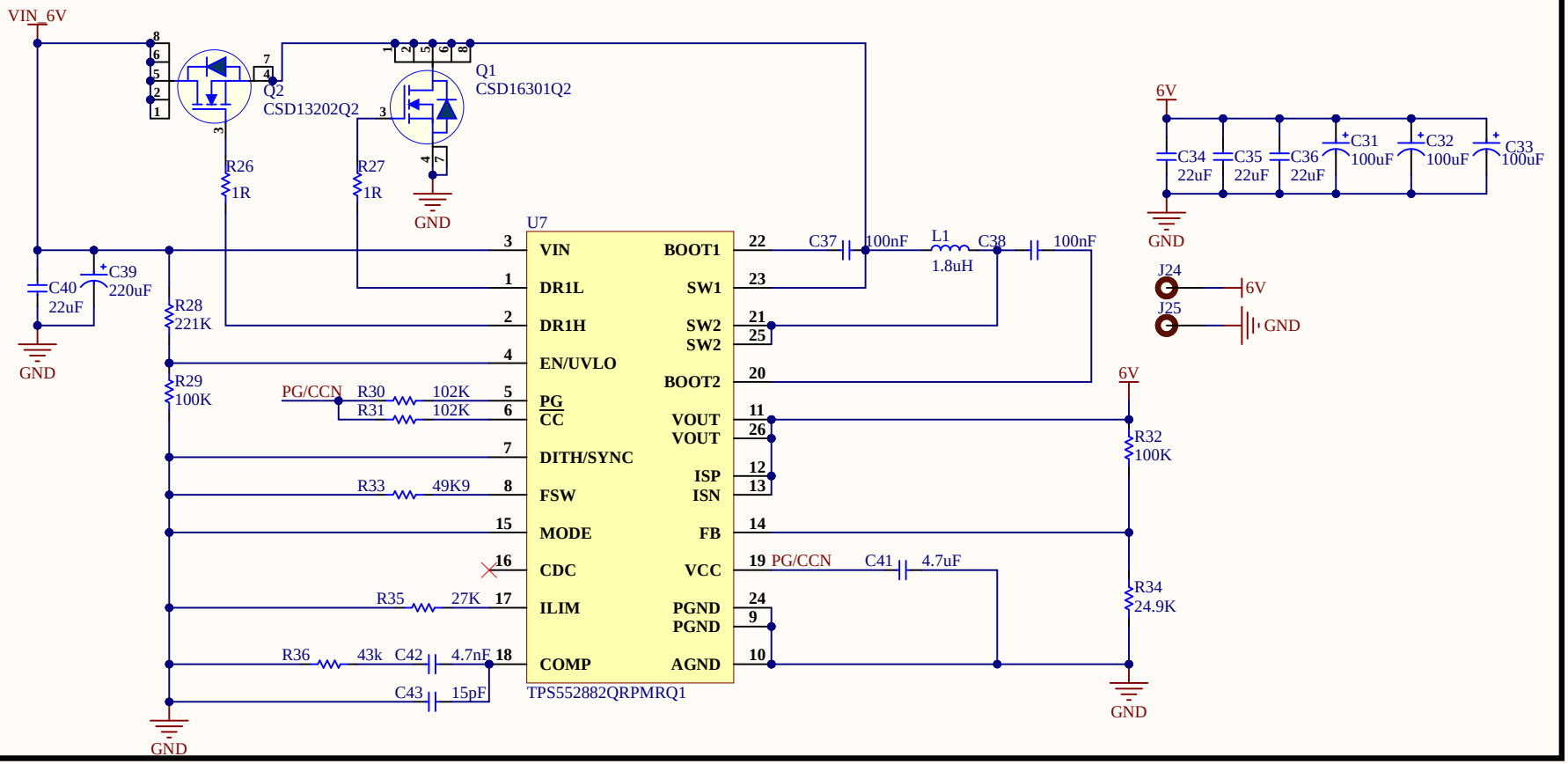
6V TO 5V CONVETER

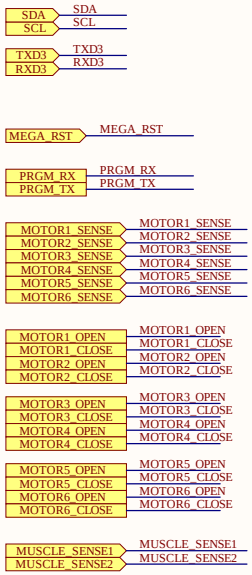


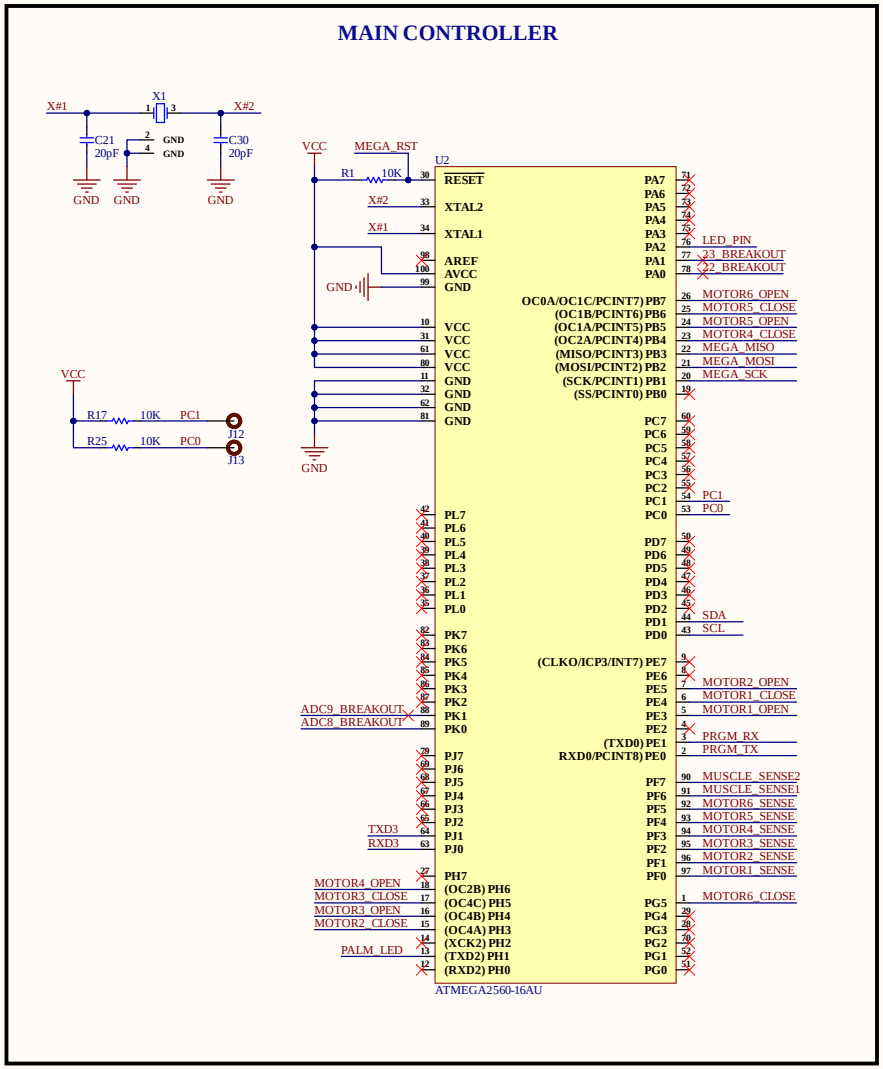
DIODE ORING

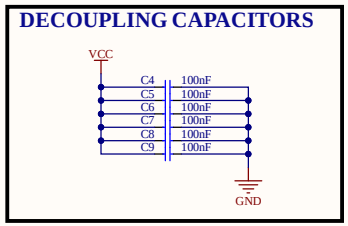


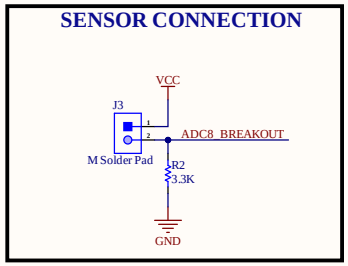
BUCK-BOOST REGULATOR

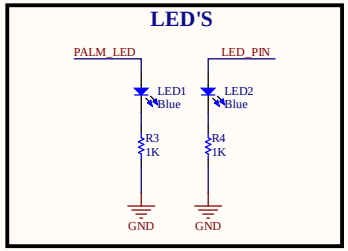


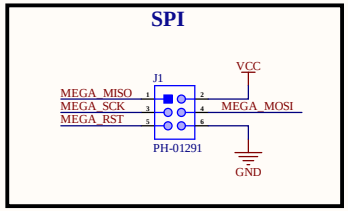

SDA
SCL
TXD3
RXD3
MEGA_RST
PRGM_RX
PRGM_TX
MOTOR1_SENSE
MOTOR2_SENSE
MOTOR3_SENSE
MOTOR4_SENSE
MOTOR5_SENSE
MOTOR6_SENSE
MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE
MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE
MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE
MUSCLE_SENSE1
MUSCLE_SENSE2


MAIN CONTROLLER
X1
X#1
X#2
C21
20pF
C30
20pF
GND
VCC
MEGA_RST
R1
10K
U2
RESET
XTAL2
XTAL1
AREF
AVCC
R17
R25
PC1
PC0
J12
J13
ATMEGA2560-16AU
LED_PIN
J3_BREAKOUT
J2_BREAKOUT
MOTOR6_OPEN
MOTOR5_CLOSE
MOTOR5_OPEN
MOTOR4_CLOSE
MEGA_MISO
MEGA_MOSI
MEGA_SCK
SDA
SCL
PRGM_RX
PRGM_TX
ADC9_BREAKOUT
ADC8_BREAKOUT
MUSCLE_SENSE2
MUSCLE_SENSE1
MOTOR6_SENSE
MOTOR5_SENSE
MOTOR4_SENSE
MOTOR3_SENSE
MOTOR2_SENSE
MOTOR1_SENSE
MOTOR6_CLOSE
MOTOR4_OPEN
MOTOR3_CLOSE
MOTOR3_OPEN
MOTOR2_CLOSE
PALM_LED


DECOUPLING CAPACITORS
VCC
C4 100nF
C5 100nF
C6 100nF
C7 100nF
C8 100nF
C9 100nF
GND


SENSOR CONNECTION
VCC
J3
M Solder Pad
ADC8_BREAKOUT
R2
3.3K
GND


LED'S
PALM_LED
LED_PIN
LED1
Blue
LED2
Blue
R3
1K
R4
1K
GND


SPI
J1
VCC
MEGA_MISO
MEGA_SCK
MEGA_RST
MEGA_MOSI
PH-01291
GND

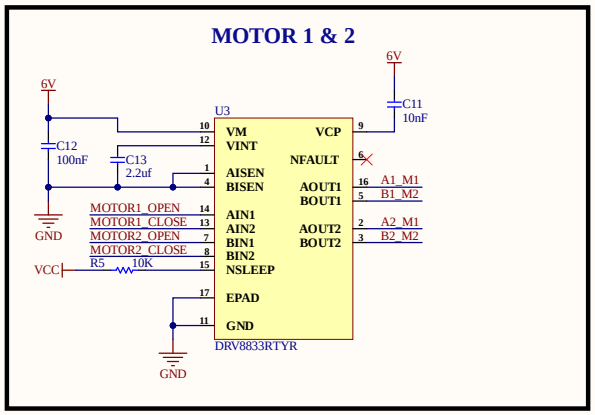
B2_M6
B1_M6
A2_M5
A1_M5
B2_M4
B1_M4
A2_M3
A1_M3
B2_M2
B1_M2
A2_M1
A1_M1

MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE

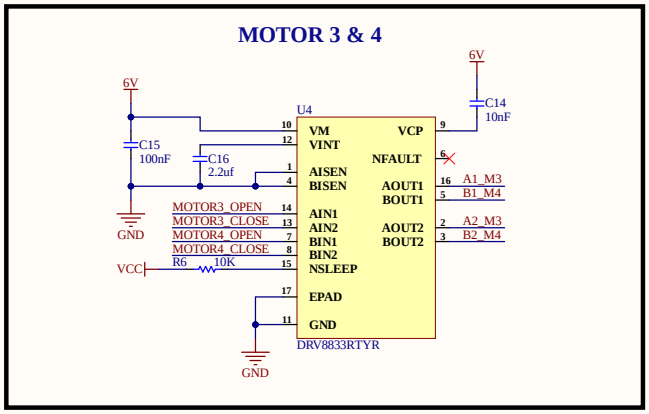
MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE

MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE

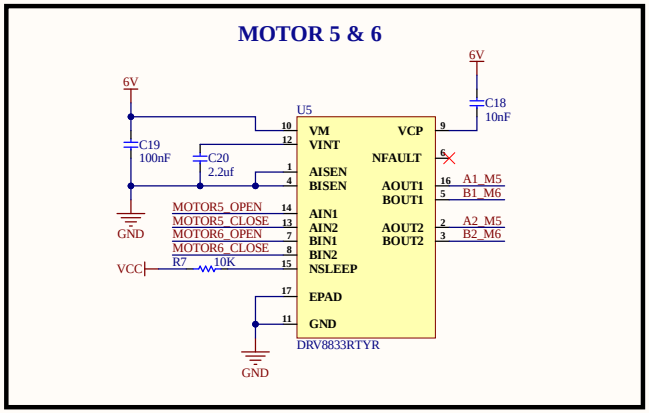
MOTOR 1 & 2

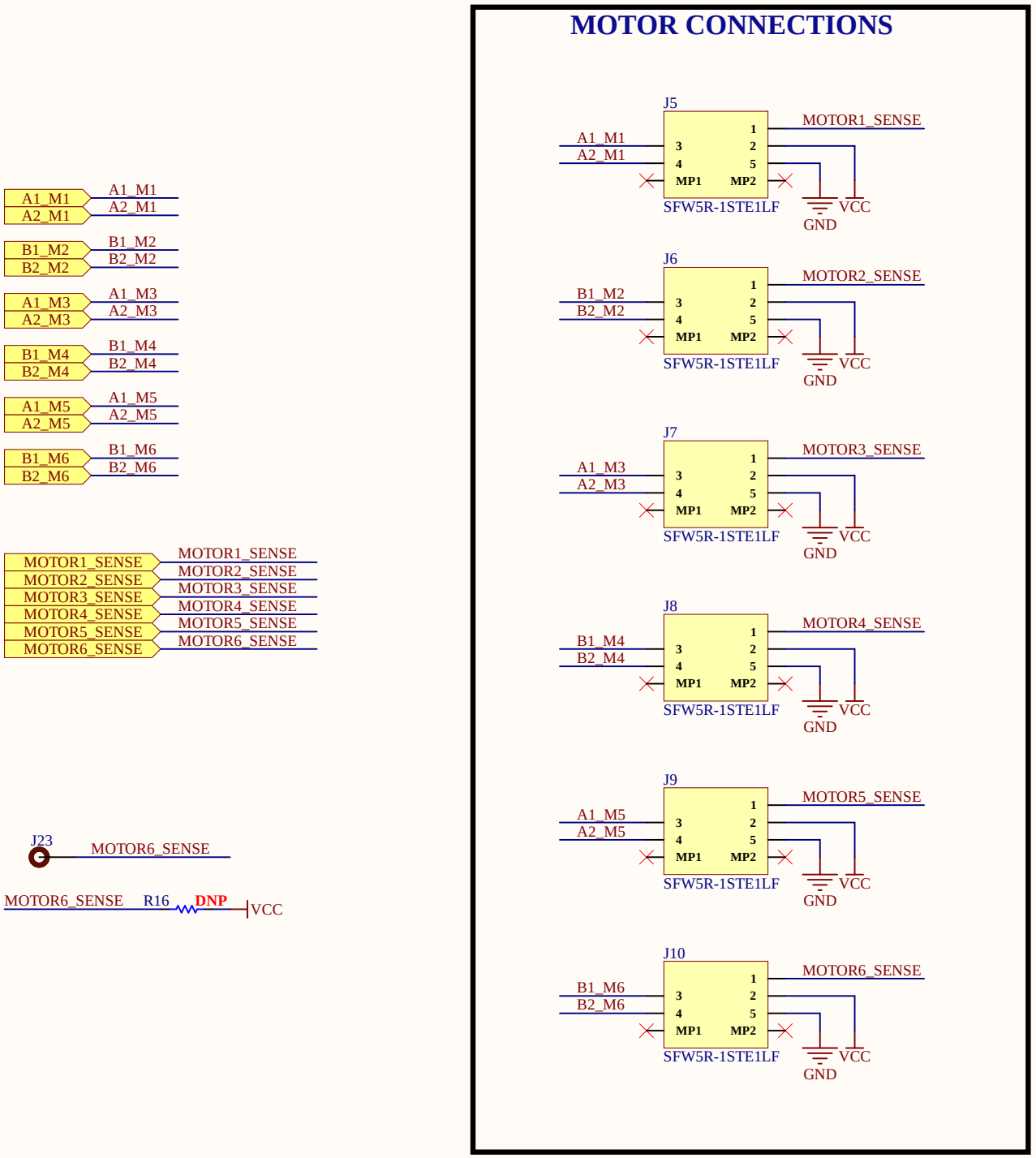


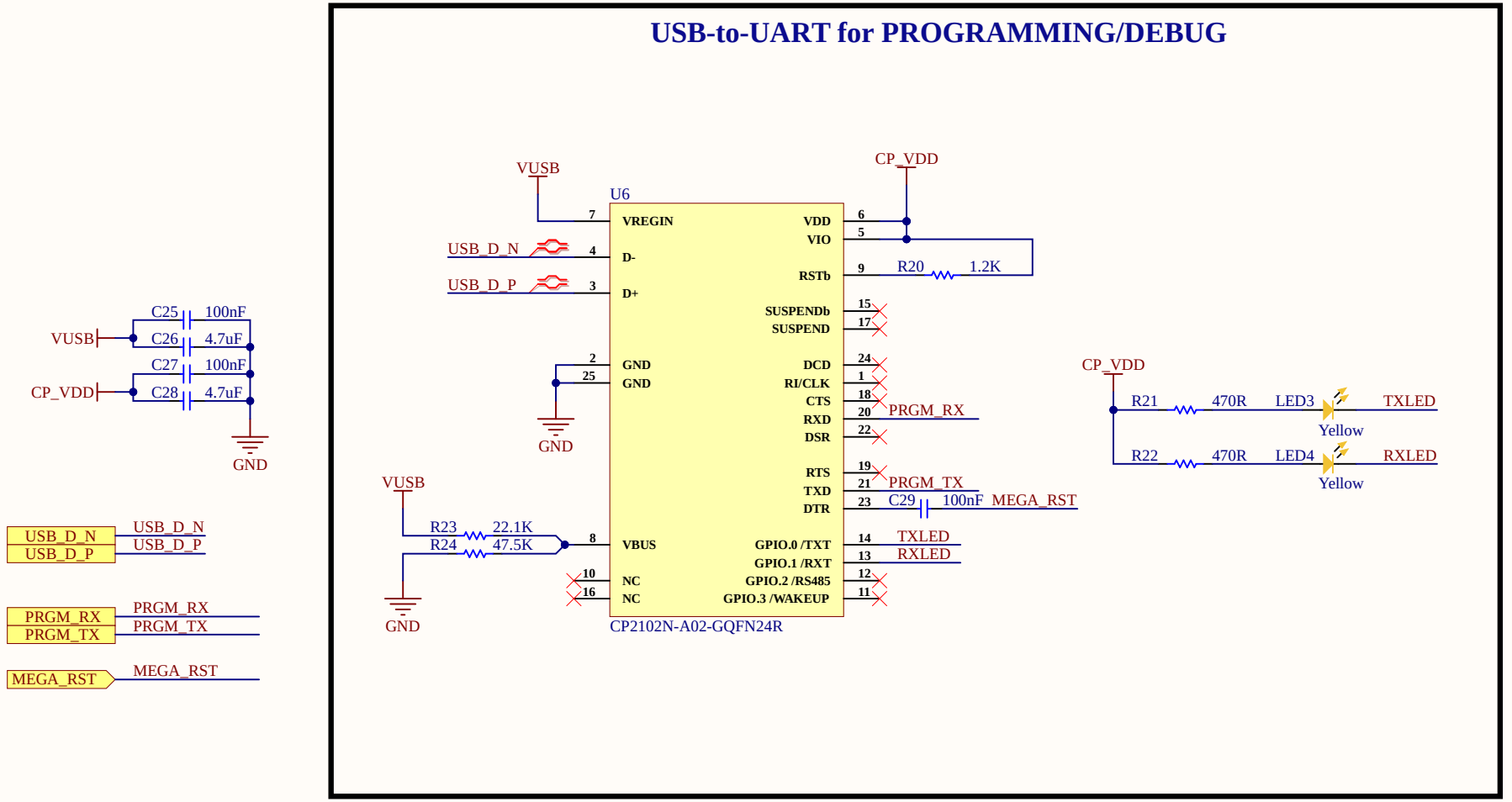
MOTOR 3 & 4



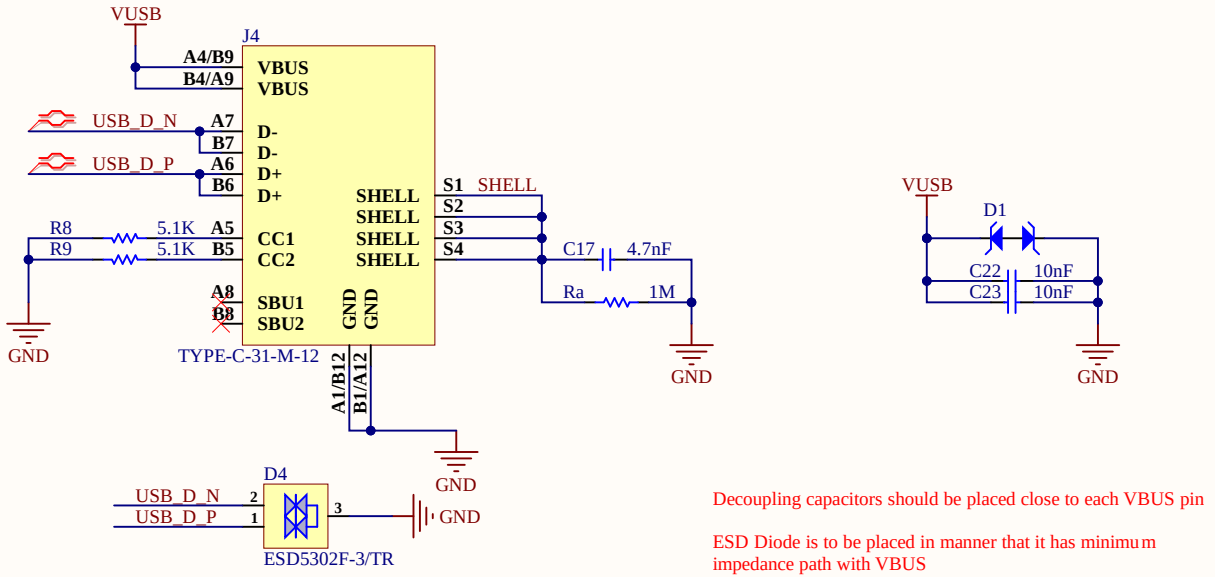
MOTOR 5 & 6




MOTOR CONNECTIONS
J5
A1_M1
A2_M1
MOTOR1_SENSE
SFW5R-1STE1LF
VCC
GND
J6
B1_M2
B2_M2
MOTOR2_SENSE
J7
A1_M3
A2_M3
MOTOR3_SENSE
J8
B1_M4
B2_M4
MOTOR4_SENSE
J9
A1_M5
A2_M5
MOTOR5_SENSE
J10
B1_M6
B2_M6
MOTOR6_SENSE
MP1
MP2
A1_M1
A2_M1
B1_M2
B2_M2
A1_M3
A2_M3
B1_M4
B2_M4
A1_M5
A2_M5
B1_M6
B2_M6
MOTOR1_SENSE
MOTOR2_SENSE
MOTOR3_SENSE
MOTOR4_SENSE
MOTOR5_SENSE
MOTOR6_SENSE
J23
MOTOR6_SENSE
R16
DNP
VCC


USB-to-UART for PROGRAMMING/DEBUG
VUSB
CP_VDD
U6
VREGIN
VDD
VIO
USB_D_N
D-
RSTb
R20
1.2K
USB_D_P
D+
SUSPENDb
SUSPEND
GND
GND
DCD
RI/CLK
CTS
RXD
DSR
PRGM_RX
RTS
TXD
DTR
PRGM_TX
C29
100nF
MEGA_RST
VUSB
R23
22.1K
R24
47.5K
VBUS
GPIO.0 /TXT
GPIO.1 /RXT
GPIO.2 /RS485
GPIO.3 /WAKEUP
TXLED
RXLED
NC
NC
GND
CP2102N-A02-GQFN24R
CP_VDD
R21
470R
LED3
TXLED
Yellow
R22
470R
LED4
RXLED
Yellow
VUSB
C25
100nF
C26
4.7uF
C27
100nF
CP_VDD
C28
4.7uF
GND
USB_D_N
USB_D_P
PRGM_RX
PRGM_TX
MEGA_RST

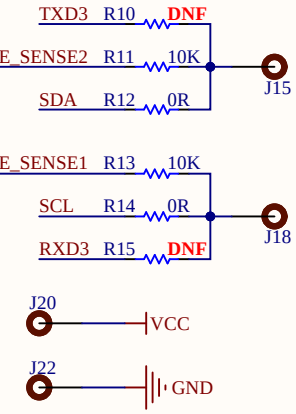
USB-C CONNECTOR

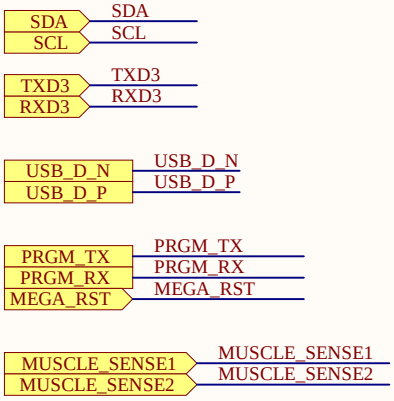


Decoupling capacitors should be placed close to each VBUS pin

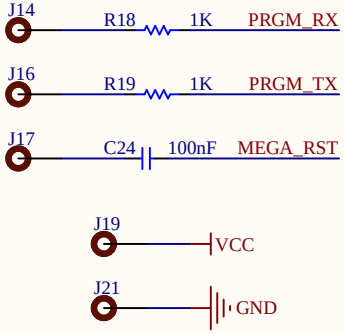
ESD Diode is to be placed in manner that it has minimum impedance path with VBUS

AUDIO CONNECTION





USB CONNECTION



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.39mil	3.5	
1	Top Layer		1.38mil		
	Dielectric 2	PP-006	8.27mil	4.1	
2	Signal_1	CF-004	0.39mil		
	Dielectric 1	FR-4	41.93mil	4.8	
3	Signal_2	CF-004	0.39mil		
	Dielectric 3	PP-006	8.27mil	4.1	
4	Bottom Layer		1.38mil		
	Bottom Solder	Solder Resist	0.39mil	3.5	
	Bottom Overlay				
Total board thickness:			62.80mil		

